

• Easy → #20 → Valid Parentheses

▲ **Problem Statement** → Given a string containing brackets, determine if all brackets are correctly matched & closed in the correct order.

▲ **Main Solution** → Use a "Stack" to store opening brackets.

- Opening Brackets → "push"
- Closing Brackets → check the top
- If they match → "pop"
- If not → return "false"

At the end, the stack must be empty.

▲ **Brute Force** → Repeatedly find matching pairs like "()", "[]" & "{ }" & remove them until no more pairs can be removed.

▲ **Algorithm** → 1] Create an empty stack

2] Loop through each character

3] If it is a ^{opening} ~~closing~~ bracket, push it.

4] If it is a closing bracket :- If the stack is empty, return "false"

• Check if it matches the top.

- If yes, pop it.

- Otherwise, return "false".

5. Return "true", if the stack is empty.

▲ Important Points: • "Stack" follows LIFO :- Last In, First out

- Push opening brackets.

- A closing bracket must match the bracket on top

- If the stack is empty when a closing bracket appears → "False"

- At the end, the stack must be empty.

▲ Complexity → Brute force → $O(n^2)$ & $O(n)$

Main Solution → $O(n)$ & $O(n)$